Lecture (4)

Nile University

Analytical Geometry & Calculus I

MTH 111

Outline

•Hyperbolic Functions

- **Hyperbolic Functions**
- **Derivatives of Hyperbolic Functions**
- **Inverse Hyperbolic Functions**
- **Derivatives of Inverse Hyperbolic Functions**

Definition of Hyperbolic Functions

Hyperbolic Sine: $$\sinh(x) = \frac{e^{x} - e^{-x}}{2}$$

Hyperbolic Cosine: $$\cosh(x) = \frac{e^{x} + e^{-x}}{2}$$

Hyperbolic Tan: $$\tanh(x) = \frac{\sinh(x)}{\cosh(x)} = \frac{e^{x} - e^{-x}}{e^{x} + e^{-x}}$$

Definition of Hyperbolic Functions

Hyperbolic Cot: $\coth(x) = \dfrac{\cosh(x)}{\sinh(x)} = \dfrac{e^{x} + e^{-x}}{e^{x} - e^{-x}}$, $x \neq 0$

Hyperbolic Sec: $\operatorname{sech}(x) = \dfrac{1}{\cosh(x)} = \dfrac{2}{e^{x} + e^{-x}}$

Hyperbolic Cosec: $\operatorname{csch}(x) = \dfrac{1}{\sinh(x)} = \dfrac{2}{e^{x} - e^{-x}}$, $x \neq 0$

Identities of Hyperbolic Functions

Prove that $\cosh x + \sinh x = e^x$

$$\cosh x - \sinh x = e^{-x}$$

Note:(use the definition for any proof)

$\cosh x + \sinh x =$ $\dfrac{e^x + e^{-x}}{2} + \dfrac{e^x - e^{-x}}{2}$

$$\frac{e^x + e^{-x} + e^x - e^{-x}}{2} = \frac{2\,e^x}{2} = e^x$$

Identities of Hyperbolic Functions

Prove that $\cosh x + \sinh x = e^x$

$$\cosh x - \sinh x = e^{-x}$$

Note:(use the definition for any proof)

$$\cosh x - \sinh x = \frac{e^x + e^{-x}}{2} - \frac{e^x - e^{-x}}{2}$$

$$\frac{e^x + e^{-x} - e^x + e^{-x}}{2} = \frac{2\,e^{-x}}{2} = e^{-x}$$

Identities of Hyperbolic Functions

Prove that $\cosh(-x) = \cosh(x)$ (even)

$\sinh(-x) = -\sinh(x)$ (odd)

$\tanh(-x) = -\tanh(x)$ (odd)

Proof

$$\cosh(x) = \frac{e^{x} + e^{-x}}{2}$$

$$\cosh(-x) = \frac{e^{-x} + e^{-(-x)}}{2} = \frac{e^{-x} + e^{x}}{2} = \cosh(x)$$

Even function

Identities of Hyperbolic Functions

Prove that $\cosh(-x) = \cosh(x)$ (even)

$\sinh(-x) = -\sinh(x)$ (odd)

$\tanh(-x) = -\tanh(x)$ (odd)

Proof

$$\sinh(x) = \frac{e^{x} - e^{-x}}{2}$$

$$\sinh(-x) = \frac{e^{-x} - e^{-(-x)}}{2} = \frac{e^{-x} - e^{x}}{2} = -\sinh(x)$$

Odd function

Identities of Hyperbolic Functions

(3) $\cosh^2 x - \sinh^2 x = 1$

$1 - \tanh^2 x = \operatorname{sech}^2 x \quad , \quad \coth^2 x - 1 = \operatorname{csch}^2 x$

Proof $\cosh^2 x - \sinh^2 x = 1$

$$\left(\frac{e^x + e^{-x}}{2}\right)^2 - \left(\frac{e^x - e^{-x}}{2}\right)^2 =$$

$$\frac{e^{2x} + 2 + e^{-2x}}{4} - \frac{e^{2x} - 2 + e^{-2x}}{4} = \frac{4}{4} = 1$$

Identities of Hyperbolic Functions

Example **Prove that**

$$\frac{1-\tanh x}{1+\tanh x} = e^{-2x}$$

$$\frac{1-\tanh x}{1+\tanh x} = \frac{1-\dfrac{\sinh x}{\cosh x}}{1+\dfrac{\sinh x}{\cosh x}}$$

$$= \frac{\cosh x-\sinh x}{\cosh x+\sinh x} = \frac{e^{-x}}{e^{x}} = e^{-2x}$$

Identities of Hyperbolic Functions

Prove that

$$\sinh 2x = 2\sinh x \cosh x$$

$$\cosh 2x = \cosh^2 x + \sinh^2 x$$

$$\cosh^2 x = \frac{1}{2}(\cosh 2x + 1)$$

$$\sinh^2 x = \frac{1}{2}(\cosh 2x - 1)$$

Try

Identities of Hyperbolic Functions

Example () Prove that

$$\ln(\cosh nx + \sinh nx)$$

$$\ln(\cosh nx + \sinh nx) = \ln e^{nx} = nx$$

Identities of Hyperbolic Functions

Example () Prove that

$$\cosh(\ln x) = \frac{x^2 + 1}{2x}$$

$$\cosh(\ln x) = \frac{e^{\ln x} + e^{-\ln x}}{2}$$

$$= \frac{e^{\ln x} + e^{\ln \frac{1}{x}}}{2} = \frac{x + \frac{1}{x}}{2} = \frac{x^2 + 1}{2x}$$

Derivatives of Hyperbolic Functions

$$\frac{d}{dx}\sinh x = \cosh x$$

$$\frac{d}{dx}\tanh x = \sec h^2 x$$

$$\frac{d}{dx}\sec hx = -\sec h\, x \tanh x$$

$$\frac{d}{dx}\cosh x = \sinh x$$

$$\frac{d}{dx}\coth x = -cs\,c\,h^2 x$$

$$\frac{d}{dx}cs\,c\,hx = -cs\,c\,h\,x \coth x$$

Derivatives of Hyperbolic Functions

Proofs

$$[\sinh x]' = \left[\frac{e^x - e^{-x}}{2}\right]'$$

$$= \frac{e^x + e^{-x}}{2} = \cosh x$$

Derivatives of Hyperbolic Functions

Proofs

$$[\tanh x]' = \left[\frac{\sinh x}{\cosh x}\right]'$$

$$= \frac{\cosh x\,(\cosh x) - \sinh x\,(\sinh x)}{\cosh^2 x}$$

$$= \frac{\cosh^2 x - \sinh^2 x}{\cosh^2 x}$$

$$= \frac{1}{\cosh^2 x} = \operatorname{sech}^2 x$$

Derivatives of Hyperbolic Functions

Proofs

$$[\operatorname{sech} x]' = \left[\frac{1}{\cosh x}\right]'$$

$$= \frac{0 - \sinh x}{\cosh^2 x}$$

$$= -\frac{1}{\cosh x} * \frac{\sinh x}{\cosh x}$$

$$= -\operatorname{sech} x \tanh x$$

Derivatives of Hyperbolic Functions (with chain rule)

$$\frac{d}{dx}\sinh u = \cosh u \frac{du}{dx}$$

$$\frac{d}{dx}\cosh u = \sinh u \frac{du}{dx}$$

$$\frac{d}{dx}\tanh u = \sec h^2 u \frac{du}{dx}$$

$$\frac{d}{dx}\coth u = -cs\,c\,h^2 u \frac{du}{dx}$$

$$\frac{d}{dx}\sec hu = -\sec h\,u \tanh u \frac{du}{dx}$$

$$\frac{d}{dx}\csc hu = -cs\,c\,h\,u \coth u \frac{du}{dx}$$

Derivatives of Hyperbolic Functions

Example Find y' if $y = \sqrt{x}\tanh x + \dfrac{1}{1+\operatorname{sech} 3x}$

$$y' = \frac{1}{2\sqrt{x}}\tanh x + \sqrt{x}\operatorname{sech}^2 x + \frac{0 - (-\operatorname{sech} 3x \tanh 3x)(3)}{(1+\operatorname{sech} 3x)^2}$$

$$= \frac{\tanh x}{2\sqrt{x}} + \sqrt{x}\operatorname{sech}^2 x + \frac{3\operatorname{sech} 3x \tanh 3x}{(1+\operatorname{sech} 3x)^2}$$

Derivatives of Hyperbolic Functions

Example () Find y' if $y = \ln(\cosh x) - e^{\sinh(x^3)}$

$$y' = \frac{\sinh x}{\cosh x} - e^{\sinh(x^3)} \cosh(x^3)\,(3x^2)$$

Derivatives of Hyperbolic Functions

Example () Find y' if $y = \dfrac{(4-x^2)*\sin^3 x}{6^{\cosh x}*\sqrt{\operatorname{csch} x}}$

$$\ln y = \ln(4-x^2) + 3\ln(\sin x) - \cosh x \ln 6 - \frac{1}{2}\ln(\operatorname{csch} x)$$

$$\frac{y'}{y} = \frac{-2x}{4-x^2} + 3\frac{\cos x}{\sin x} - (\ln 6)\sinh x - \frac{1}{2} * \frac{-\operatorname{csch} x \coth x}{\operatorname{csch} x}$$

$$y' = y\left[\frac{-2x}{4-x^2} + 3\cot x - (\ln 6)\sinh x + \frac{1}{2}\coth x\right]$$

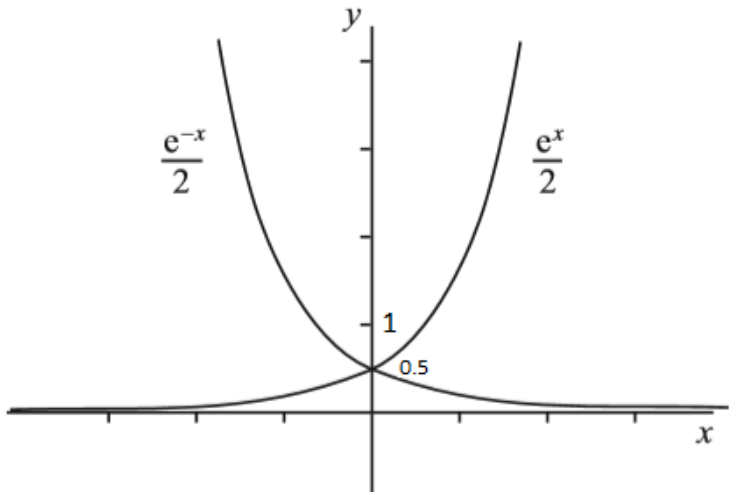
Find y' if $y = x^x \tanh x$ (try to solve)

Graphs of Hyperbolic Functions

Hyperbolic Cos:

$$\cosh(x) = \frac{e^x + e^{-x}}{2}$$

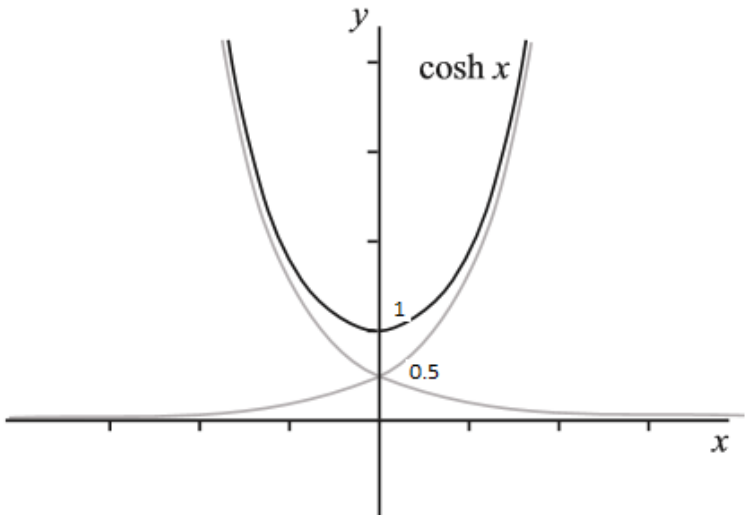
$$\cosh(x) = \frac{e^x}{2} + \frac{e^{-x}}{2}$$



$$Domain = R$$

$$Range = [1, \infty)$$

Even function

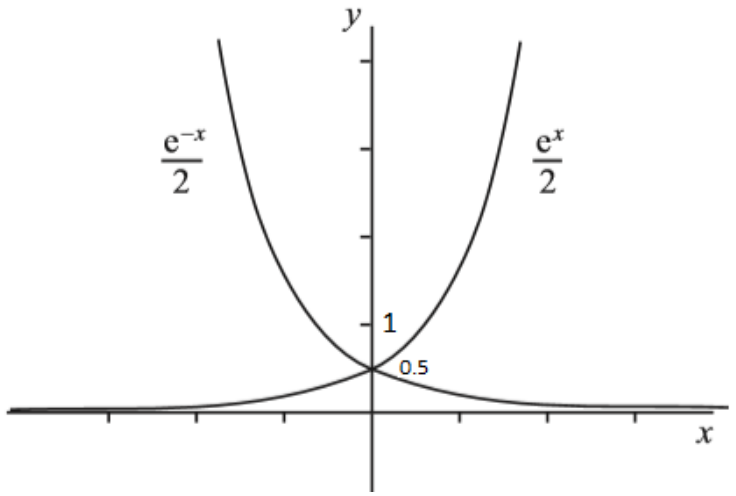


Graphs of Hyperbolic Functions

Hyperbolic Sin:

$$\sinh(x) = \frac{e^{x} - e^{-x}}{2}$$

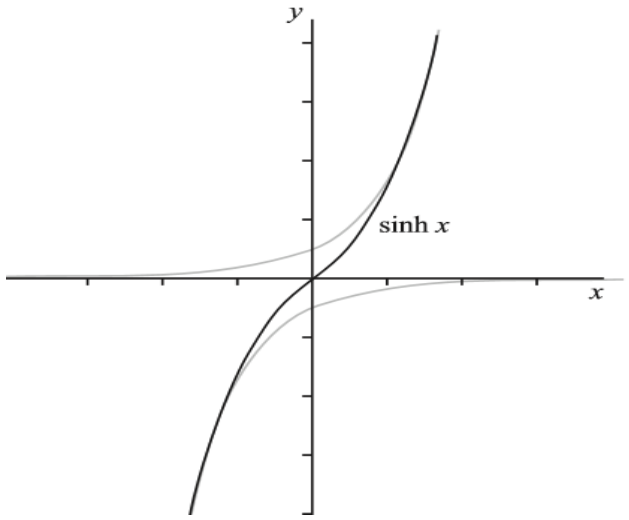
$$\sinh(x) = \frac{e^{x}}{2} - \frac{e^{-x}}{2}$$



$$Domain = R$$

$$Range = R$$

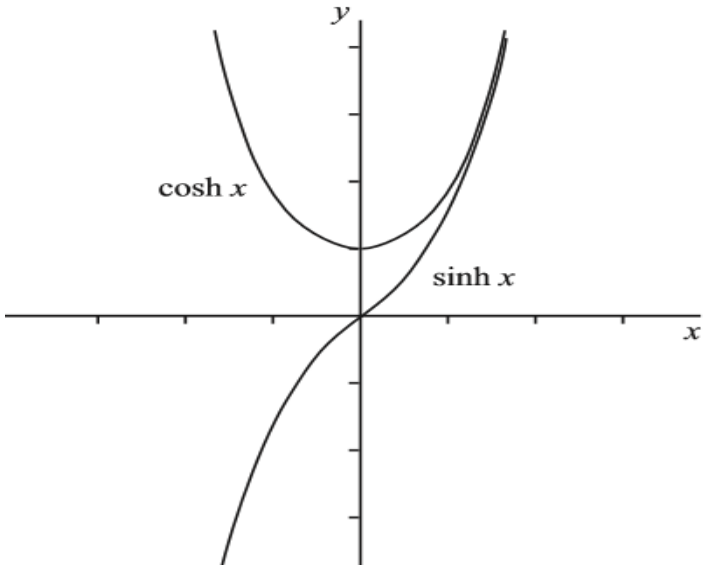
Odd Function



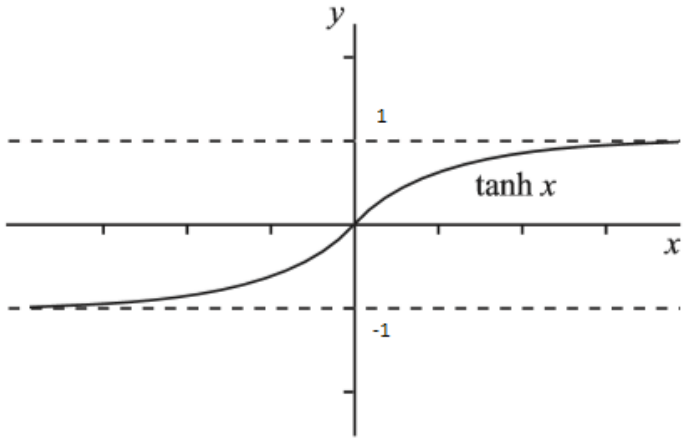
Graphs of Hyperbolic Functions

$$\tanh(x) = \frac{\sinh(x)}{\cosh(x)} = \frac{e^{x} - e^{-x}}{e^{x} + e^{-x}}$$

If $x > 0$, $\tanh(x)$ is positive
If $x < 0$, $\tanh(x)$ is negative
If $x = 0$ $\tanh(x) = 0$



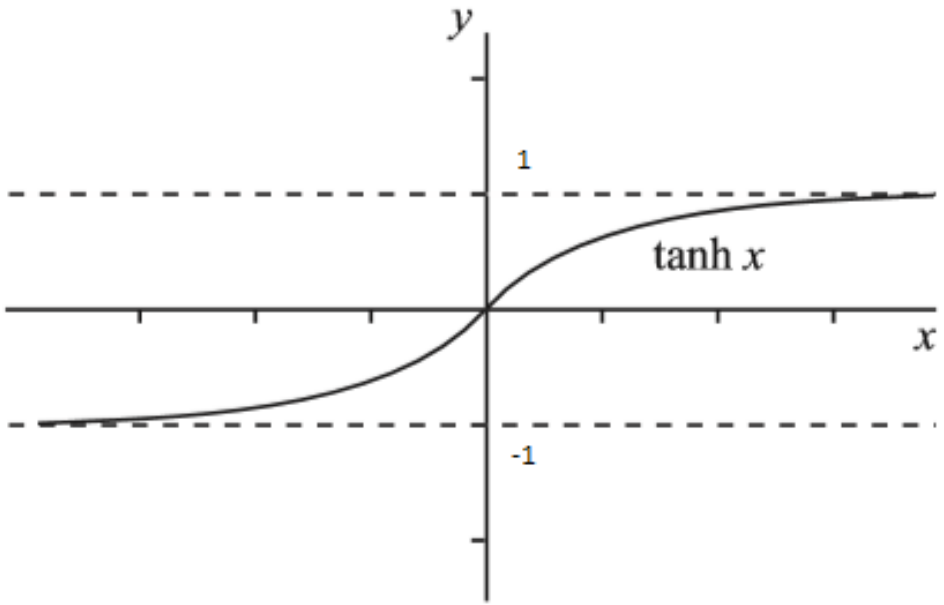
$$|\sinh x| \prec \cosh x \Rightarrow \quad |\tanh x| \prec 1$$



$$\lim_{x \to \infty} \frac{e^{x} - e^{-x}}{e^{x} + e^{-x}} = 1$$

Graphs of Hyperbolic Functions

$$\tanh(x) = \frac{\sinh(x)}{\cosh(x)} = \frac{e^{x} - e^{-x}}{e^{x} + e^{-x}}$$



$$Domain = R$$

$$Range = (-1,1)$$

Odd function

Graphs of Hyperbolic Functions

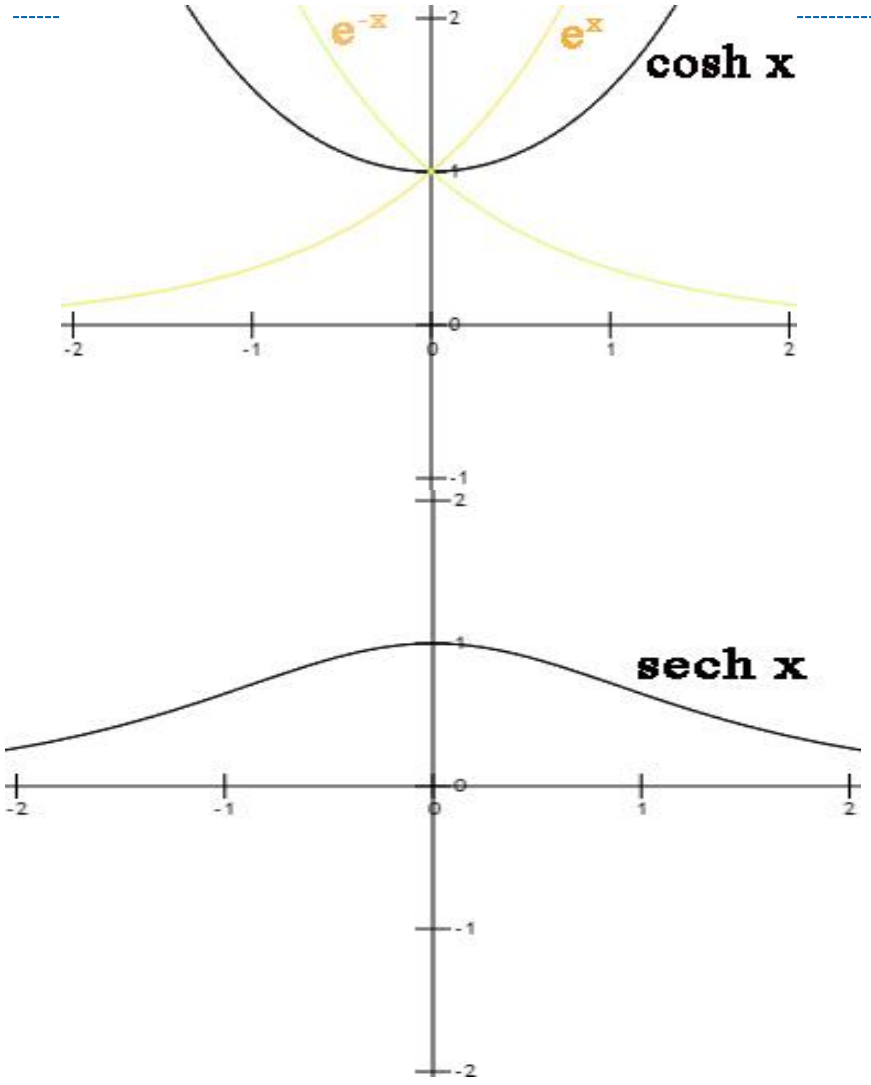
$$\operatorname{sech}(x) = \frac{1}{\cosh(x)} = \frac{2}{e^{x} + e^{-x}}$$

Reciprocal

$$Domain = R$$

$$Range = (\,0\,,1\,]$$

Even function



Graphs of Hyperbolic Functions

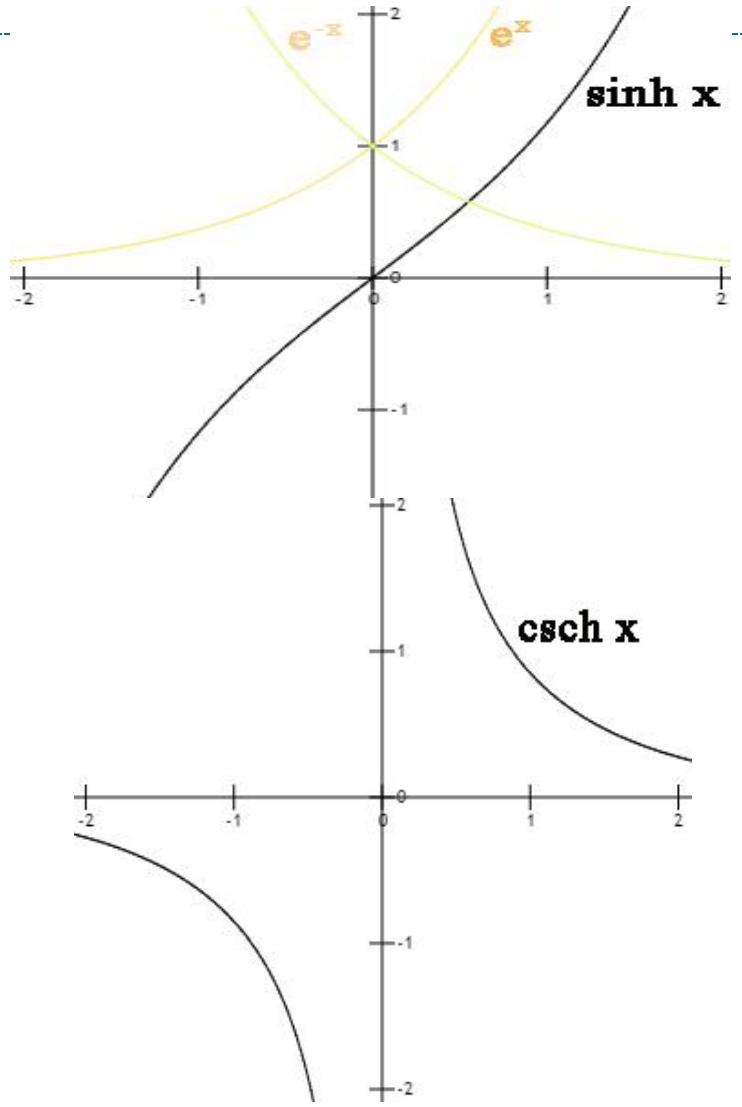
$$cs\,\mathrm{ch}(x) = \frac{1}{\sinh(x)} = \frac{2}{e^{x} - e^{-x}}$$

Reciprocal

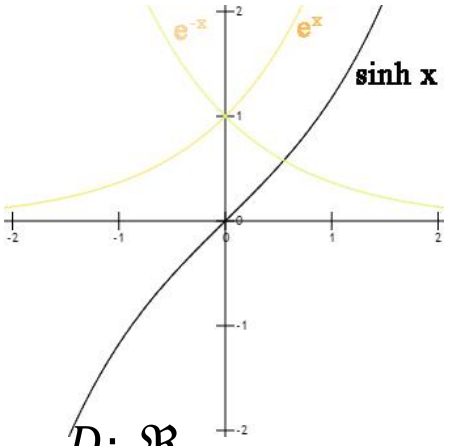
$$Domain = R - \{0\}$$

$$Range = R - \{0\}$$

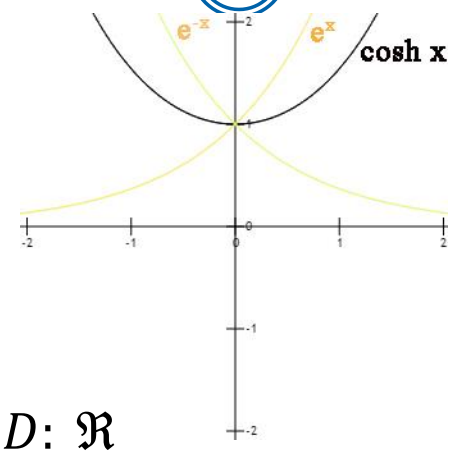
Odd function



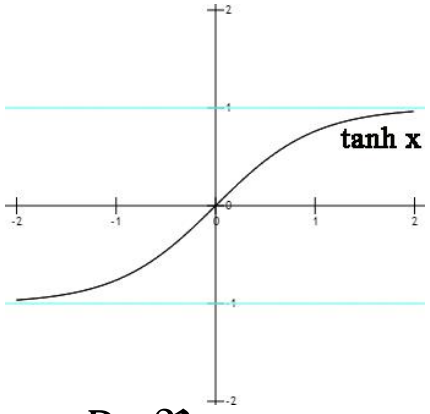
Graphs of Hyperbolic Functions



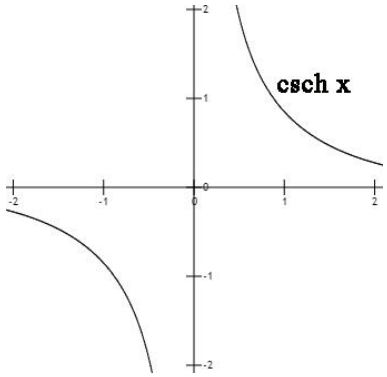
D: $\Re$
R: $\Re$



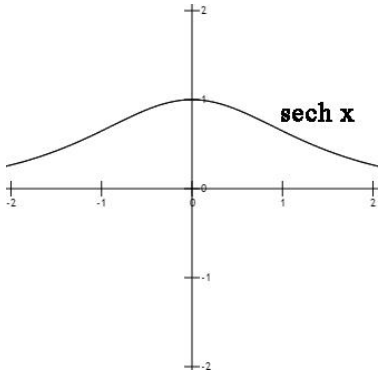
D: $\Re$
R: $[1, \infty)$



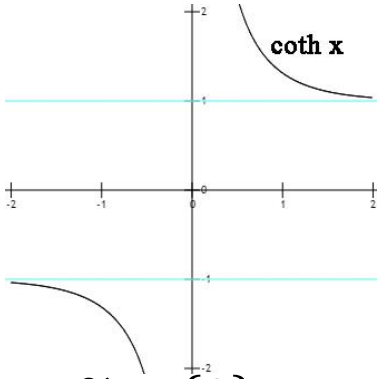
D: $\Re$
R: $(-1,1)$



D: $\Re - \{0\}$
R: $\Re - \{0\}$



D: $\Re$
R: $(0, 1]$



D: $\Re - \{0\}$
R: $\Re - [-1,1]$

Inverse Hyperbolic Functions

- $y = \sinh^{-1} x \quad \Leftrightarrow \quad x = \sinh y$
- $y = \cosh^{-1} x \quad \Leftrightarrow \quad x = \cosh y \,, \qquad 0 \leq y$
- $y = \tanh^{-1} x \quad \Leftrightarrow \quad x = \tanh y$
- $y = \coth^{-1} x \quad \Leftrightarrow \quad x = \coth y$
- $y = \operatorname{sech}^{-1} x \quad \Leftrightarrow \quad x = \operatorname{sech} y \,, \qquad 0 \leq y$
- $y = \operatorname{csch}^{-1} x \quad \Leftrightarrow \quad x = \operatorname{csch} y$

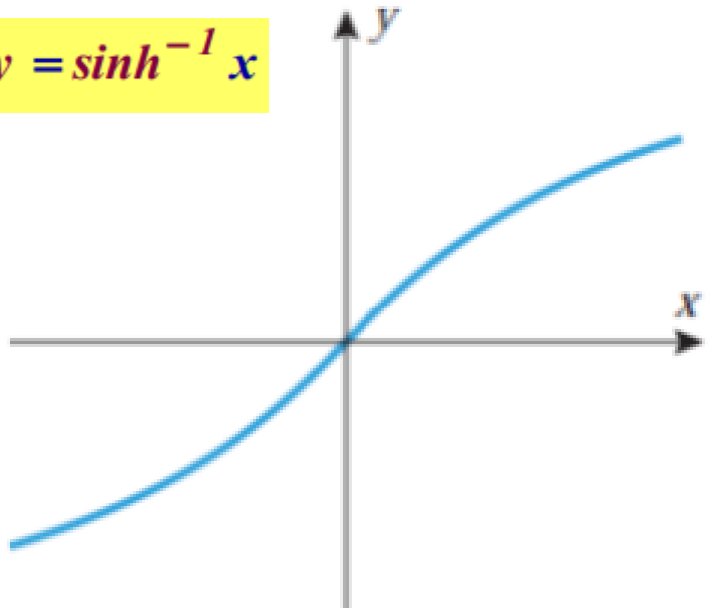
Graphs of Inverse Hyperbolic Functions

$y = \sinh x$



one to one

$y = \sinh^{-1} x$



$Domain = \mathbb{R}$

$Range = \mathbb{R}$

Odd function

$\sinh(-x) = -\sinh x$

$Domain = \mathbb{R}$

$Range = \mathbb{R}$

Odd function

$\sinh^{-1}(-x) = -\sinh^{-1} x$

Graphs of Inverse Hyperbolic Functions

$y = \cosh x$

Not one to one



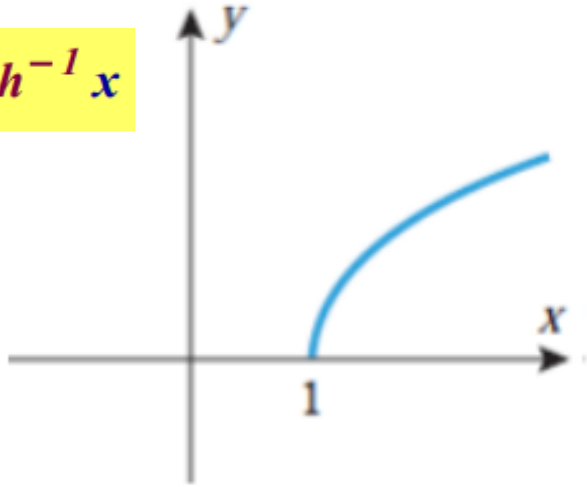
$\text{Domain} = \mathbb{R}$

$\text{Domain} = [0, \infty)$

$\text{Range} = [1, \infty)$

$\cosh(-x) = \cosh x$

$y = \cosh^{-1} x$



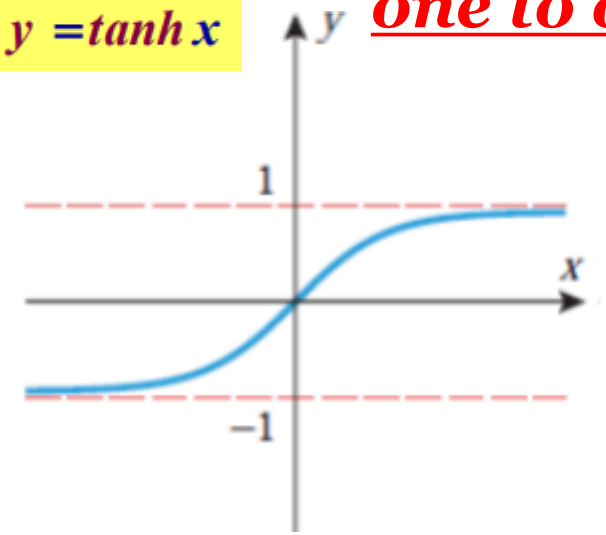
$\text{Domain} = [1, \infty)$

$\text{Range} = [0, \infty)$

Neither even nor odd function

Graphs of Inverse Hyperbolic Functions

$y = \tanh x$ ***one to one***



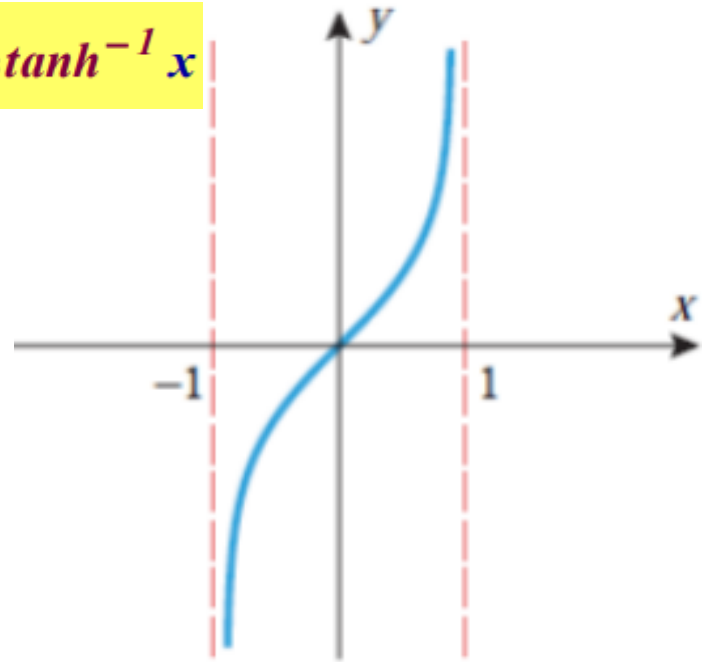
$Domain = \mathbb{R}$

$Range = (-1, 1)$

Odd function

$\tanh(-x) = -\tanh x$

$y = \tanh^{-1} x$



$Domain = (-1, 1)$

$Range = \mathbb{R}$

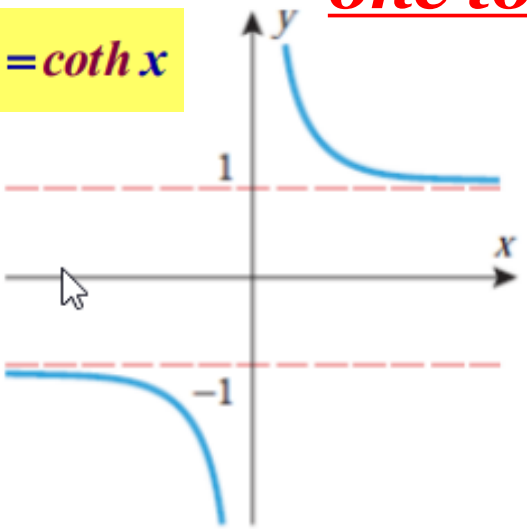
Odd function

$\tanh^{-1}(-x) = -\tanh^{-1} x$

Graphs of Inverse Hyperbolic Functions

one to one

$y = \coth x$



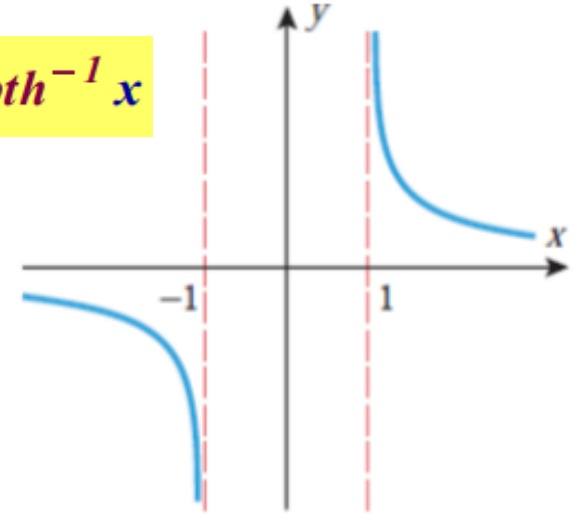
$Domain = \mathbb{R} - \{0\}$

$Range = \mathbb{R} - [-1, 1]$

Odd function

$\coth(-x) = -\coth x$

$y = \coth^{-1} x$



$Domain = \mathbb{R} - [-1, 1]$

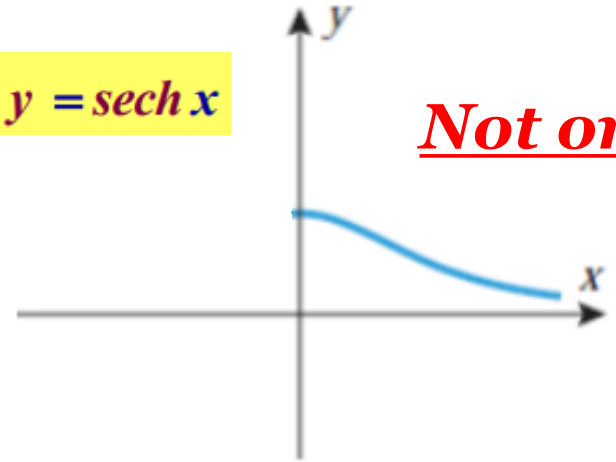
$Range = \mathbb{R} - \{0\}$

Odd function

$\coth^{-1}(-x) = -\coth^{-1} x$

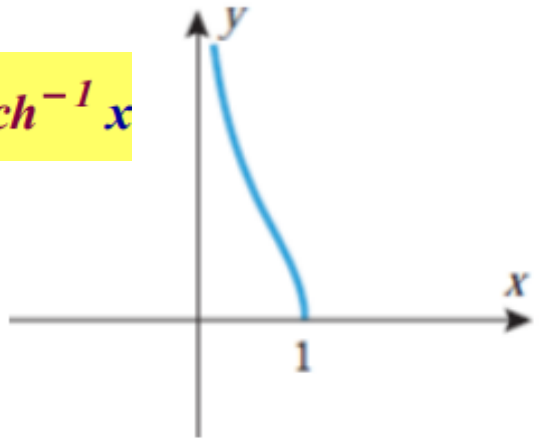
Graphs of Inverse Hyperbolic Functions

$y = \operatorname{sech} x$



Not one to one

$y = \operatorname{sech}^{-1} x$



$\text{Domain} = \mathbb{R}$

$\text{Domain} = [0, \infty)$

$\text{Range} = (0, 1]$

$\operatorname{sech}(-x) = \operatorname{sech} x$

$\text{Domain} = (0, 1]$

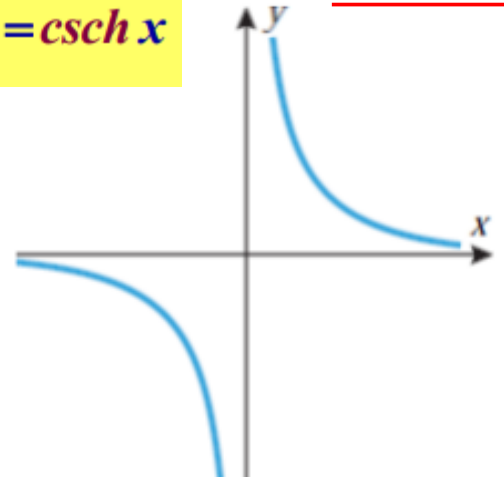
$\text{Range} = [0, \infty)$

Neither even nor odd function

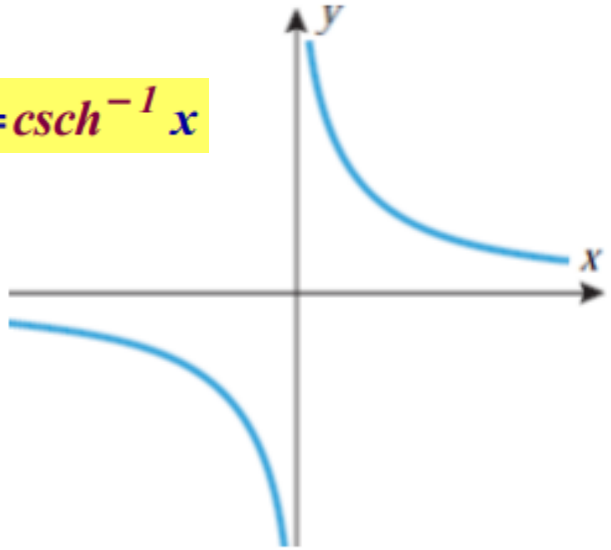
Graphs of Inverse Hyperbolic Functions

one to one

$y = \operatorname{csch} x$



$y = \operatorname{csch}^{-1} x$



$Domain = \mathbb{R} - \{0\}$

$Range = \mathbb{R} - \{0\}$

Odd function

$\operatorname{csch}(-x) = -\operatorname{csch} x$

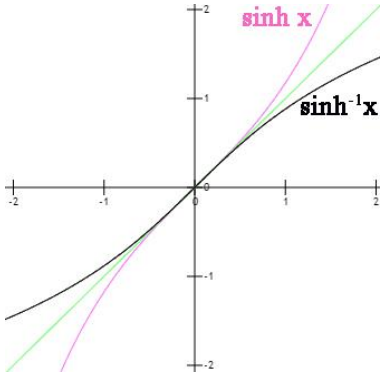
$Domain = \mathbb{R} - \{0\}$

$Range = \mathbb{R} - \{0\}$

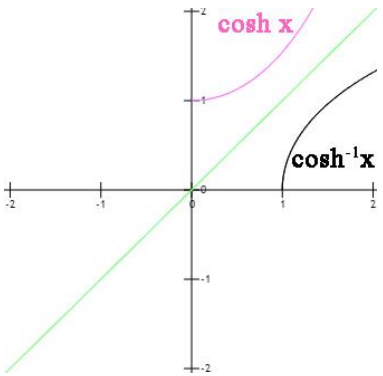
Odd function

$\operatorname{csch}^{-1}(-x) = -\operatorname{csch}^{-1} x$

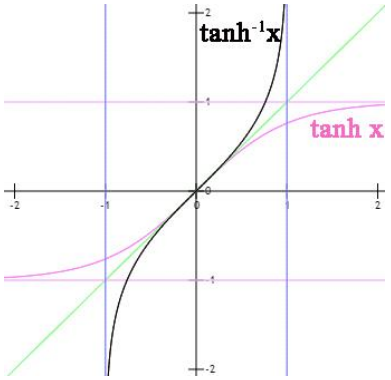
Graphs of Inverse Hyperbolic Functions



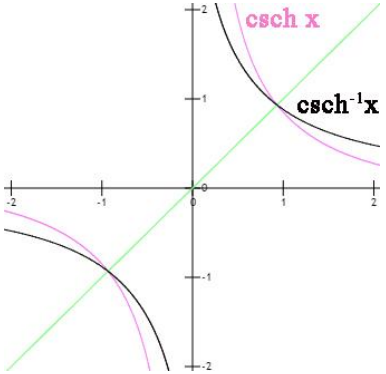
D: $\Re$
R: $\Re$



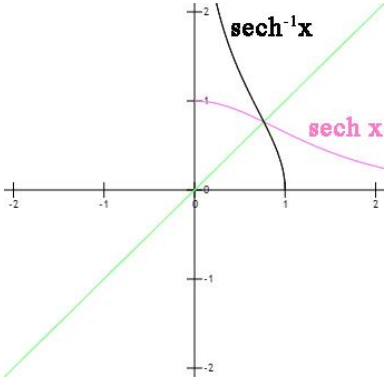
D: $[1, \infty)$
R: $[0, \infty)$



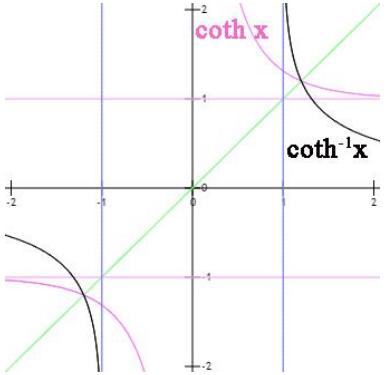
D: $(-1,1)$
R: $\Re$



D: $\Re - \{0\}$
R: $\Re - \{0\}$



D: $(0, 1]$
R: $[0, \infty)$



D: $\Re - [-1,1]$
R: $\Re - \{0\}$

Derivatives of Inverse Hyperbolic Functions

$$\frac{d}{dx}\sinh^{-1}x=\frac{1}{\sqrt{1+x^2}}$$

$$\frac{d}{dx}\cosh^{-1}x=\frac{1}{\sqrt{x^2-1}}$$

$$\frac{d}{dx}\tanh^{-1}x=\frac{1}{1-x^2}$$

$$\frac{d}{dx}\coth^{-1}x=\frac{1}{1-x^2}$$

$$\frac{d}{dx}\operatorname{sech}^{-1}x=-\frac{1}{x\sqrt{1-x^2}}$$

$$\frac{d}{dx}\operatorname{csch}^{-1}x=-\frac{1}{|x|\sqrt{1+x^2}}$$

Derivatives of Inverse Hyperbolic Functions

Proofs

(i) $y = \sinh^{-1} x \quad \Rightarrow \quad \sinh y = x$

$\cosh y \;\; y' = 1$

$$y' = \frac{1}{\cosh y} = \frac{1}{\sqrt{1+\sinh^2 y}} = \frac{1}{\sqrt{1+x^2}}$$

Derivatives of Inverse Hyperbolic Functions

Proofs

(ii) $y = \tanh^{-1} x \quad \Rightarrow \quad \tanh y = x$

$\operatorname{sech}^2 y \;\; y' = 1$

$$y' = \frac{1}{\operatorname{sech}^2 y} = \frac{1}{1-\tanh^2 y} = \frac{1}{1-x^2}$$

Derivatives of Inverse Hyperbolic Functions

Proofs

(iii) $y = \operatorname{sech}^{-1} x \quad \Rightarrow \quad \operatorname{sech} y = x$

$-\operatorname{sech} y \tanh y \;\; y' = 1$

$$y' = \frac{-1}{\operatorname{sech} y \tanh y} = \frac{-1}{\operatorname{sech} y \sqrt{1-\operatorname{sech}^2 y}}$$

$$= \frac{-1}{x\sqrt{1-x^2}}$$

Derivatives of Inverse Hyperbolic Functions

Find y' if

$$\boldsymbol{y = e^x \tanh^{-1} x + 3^x \cosh^{-1} x}$$

$$y' = e^x \tanh^{-1} x + \frac{e^x}{1 - x^2}$$

$$+3^x \ln 3 \cosh^{-1} x + 3^x \frac{1}{\sqrt{x^2 - 1}}$$

Derivatives of Inverse Hyperbolic Functions

Find y'

if $y = \sin^{-1} x + x \operatorname{sech}^{-1} x + \sqrt{1 - x^2}$

$$y' = \frac{1}{\sqrt{1-x^2}} + \operatorname{sech}^{-1} x + x * \frac{-1}{x\sqrt{1-x^2}} + \frac{-2x}{2\sqrt{1-x^2}}$$

$$= \operatorname{sech}^{-1} x - \frac{x}{\sqrt{1-x^2}}$$

Show that $y'' = \csc x \cot x - \operatorname{sech} x \tanh x$ if $y = \tan^{-1}(\sinh x) + \operatorname{sech}^{-1}(\sin x)$

$$y' = \frac{1}{1 + \sinh^2 x}\cosh x + \frac{-1}{\sin x\sqrt{1 - \sin^2 x}}\cos x$$

$$= \frac{\cosh x}{\cosh^2 x} - \frac{\cos x}{\sin x \cos x}$$

$$= \frac{1}{\cosh x} - \frac{1}{\sin x} = \operatorname{sech} x - \csc x$$

$$y'' = -\operatorname{sech} x \tanh x - (-\csc x \cot x)$$

$$= \csc x \cot x - \operatorname{sech} x \tanh x$$

Logarithmic Forms of Inverse Hyperbolic Functions

Example **Prove that** $\tanh^{-1} x = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right)$

$$y = \tanh^{-1} x \quad \Rightarrow \quad \tanh y = x \quad \Rightarrow$$

$$\frac{e^y - e^{-y}}{e^y + e^{-y}} = x \quad \Rightarrow \quad e^y - e^{-y} = x(e^y + e^{-y})$$

$$e^y(1-x) = e^{-y}(1+x)$$

$$e^{2y}(1-x) = (1+x) \quad \Rightarrow$$

$$e^{2y} = \frac{1+x}{1-x} \quad \Rightarrow \quad y = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right)$$

$$\tanh^{-1} x = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right)$$

Logarithmic Forms of Inverse Hyperbolic Functions

Example: **Prove that $\sinh^{-1} x = \ln\left(x + \sqrt{x^2+1}\right)$**

$y = \sinh^{-1} x \quad \Rightarrow \quad \sinh y = x \quad \Rightarrow$

$\frac{e^y - e^{-y}}{2} = x \quad \Rightarrow \quad e^y - e^{-y} = 2x \quad \Rightarrow$

$e^{2y} - 1 = 2x\, e^y \quad \Rightarrow \quad e^{2y} - 2xe^y - 1 = 0$

$$e^y = \frac{2x \pm \sqrt{4x^2+4}}{2} = x \pm \sqrt{x^2+1}$$

$e^y = x + \sqrt{x^2+1}$ (e^y is always positive)

$y = \ln\left(x + \sqrt{x^2+1}\right)$

$\sinh^{-1} x = \ln\left(x + \sqrt{x^2+1}\right)$

Logarithmic Forms of Inverse Hyperbolic Functions

- $\sinh^{-1} x = \ln\left(x + \sqrt{x^2 + 1}\right)$
- $\cosh^{-1} x = \ln\left(x + \sqrt{x^2 - 1}\right)$, $1 \leq x$
- $\tanh^{-1} x = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right)$, $|x| < 1$
- $\coth^{-1} x = \frac{1}{2}\ln\left(\frac{x+1}{x-1}\right)$, $1 < |x|$
- $\operatorname{sech}^{-1} x = \ln\left(\frac{1+\sqrt{1-x^2}}{x}\right)$, $0 < x \leq 1$
- $\operatorname{csch}^{-1} x = \ln\left(\frac{1}{x} + \frac{\sqrt{1+x^2}}{|x|}\right)$, $x \neq 0$



See you next week

11.2 Applications of Hyperbolic Functions



Inverse Hyperbolic Identities

Odd Functions

$$\sinh^{-1}(-x) = -\sinh^{-1} x$$

$$\tanh^{-1}(-x) = -\tanh^{-1} x$$

$$\coth^{-1}(-x) = -\coth^{-1} x$$

$$\operatorname{csch}^{-1}(-x) = -\operatorname{csch}^{-1} x$$

Inverse Hyperbolic Identities

$$\operatorname{csch}^{-1} x = \sinh^{-1}\left(\frac{1}{x}\right)$$

$$\operatorname{sech}^{-1} x = \cosh^{-1}\left(\frac{1}{x}\right)$$

$$\coth^{-1} x = \tanh^{-1}\left(\frac{1}{x}\right)$$

Proof

$$y = \operatorname{csch}^{-1} x \quad \Rightarrow \quad \operatorname{csch} y = x$$

$$\sinh y = \frac{1}{x} \quad \Rightarrow \quad y = \sinh^{-1}\frac{1}{x}$$

$$\Rightarrow \quad \operatorname{csch}^{-1} x = \sinh^{-1}\frac{1}{x}$$